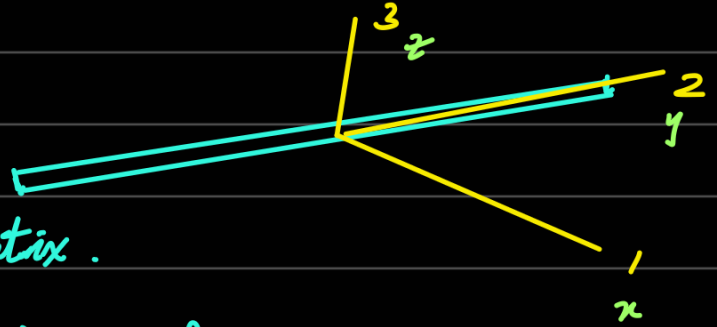


1. Moments of inertia of rigid bodies: Determine the principal moments of inertia for the following homogeneous bodies: (a) a thin rod of length l , (b) a sphere of radius R , (c) a circular cylinder of radius R and height h , and (d) a rectangular parallelepiped of sides a , b and c .

$$I_{ik} = \int dm (x_l^2 \delta_{ik} - x_i x_k)$$



For principal axes, I is a diagonal matrix.

$$\begin{aligned} \text{a) } I_{11} &= \int dm \left((x^2 + y^2 + z^2) - x^2 \right) = \int dm (y^2 + z^2) \\ &\Rightarrow I_{11} = \int_{-\frac{l}{2}}^{\frac{l}{2}} (\lambda dy) (y^2) = \frac{\lambda l^3}{12} = \boxed{\frac{Ml^2}{12}} \end{aligned}$$

$$I_{22} = \int dm \left((x^2 + y^2 + z^2) - y^2 \right) = \int dm (x^2 + z^2) = \boxed{0}$$

$$\begin{aligned} I_{33} &= \int dm \left((x^2 + y^2 + z^2) - z^2 \right) = \int dm (x^2 + y^2) \\ &\Rightarrow I_{33} = \int_{-\frac{l}{2}}^{\frac{l}{2}} \lambda dy y^2 = \boxed{\frac{Ml^2}{12}} \end{aligned}$$

b) By symmetry all moments are equal.

$$\begin{aligned} I_{11} + I_{22} + I_{33} &= \int dm \left(3(x^2 + y^2 + z^2) - (x^2 + y^2 + z^2) \right) \\ &\Rightarrow 3I = 2 \int dm r^2 \end{aligned}$$

$$\begin{aligned} \text{or } I &= \frac{2}{3} \int \rho 4\pi r^4 dr = \frac{8\pi \rho R^5}{15} \\ &= \frac{8\pi R^5}{15} \frac{M}{\frac{4}{3}\pi R^3} = \boxed{\frac{2}{5} MR^2} \end{aligned}$$

$$c) I_{11} = \int dm (x^2 + y^2 + z^2) - x^2$$

$$I_{11} = \int dm (y^2 + z^2)$$

$$y = s \sin \phi \quad z = z$$

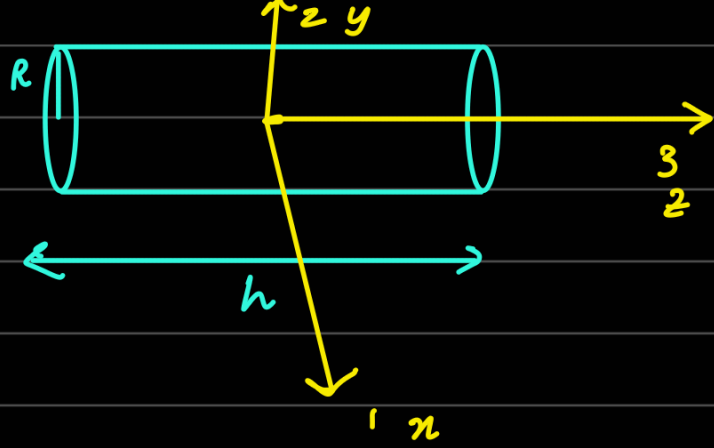
$$dm = \rho s ds d\phi dz$$

$$I_{11} = \int \rho (s^2 \sin^2 \phi + z^2) s ds d\phi dz$$

$$= \int \rho s^3 \sin^2 \phi ds d\phi dz + \int \rho z^2 s ds d\phi dz$$

$$= \rho \frac{R^4}{4} \pi H + \rho \frac{R^2}{2} 2\pi \frac{H^3}{12}$$

$$= M \left(\frac{R^2}{4} + \frac{H^2}{12} \right)$$



$$I_{22} = \int dm (x^2 + z^2) = I_{11} \text{ by symmetry.}$$

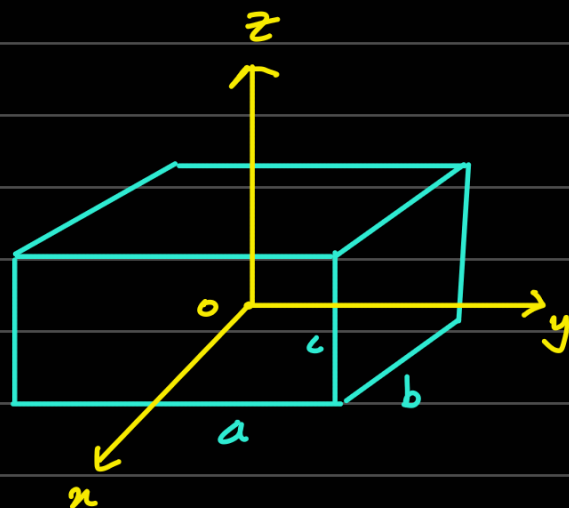
$$I_{33} = \int dm (x^2 + y^2) = \int \rho s^3 ds d\phi dz = \rho \frac{R^4}{4} 2\pi H = \frac{\rho \pi H R^4}{2}$$

$$I_{33} = \frac{MR^2}{2}$$

$$d) I_{11} = \int dm (y^2 + z^2)$$

$$= \rho \int_{-\frac{b}{2}}^{\frac{b}{2}} \int_{-\frac{c}{2}}^{\frac{c}{2}} \int_{-\frac{a}{2}}^{\frac{a}{2}} (y^2 + z^2) dy dz dx$$

$$\rho = \frac{M}{abc}$$



2. Small oscillations of a compound pendulum: Determine the frequency of small oscillations of a compound pendulum, i.e. a rigid body swinging about a fixed horizontal axis in a gravitational field.

let distance of CoM from point of suspension be l .

$$U = mgl(1 - \cos\phi)$$

$$\Omega_1 = \Omega \cos\alpha$$

$$\Omega_2 = \Omega \cos\beta \quad \Omega_3 = \Omega \cos\gamma$$

$$T = \frac{1}{2} I_1 \Omega_1^2 + \frac{1}{2} I_2 \Omega_2^2 + \frac{1}{2} I_3 \Omega_3^2 + \frac{1}{2} m l^2 \dot{\Omega}^2 \quad \bar{\Omega} = \dot{\phi}$$

$$T = \frac{\Omega^2}{2} (I_1 \cos^2\alpha + I_2 \cos^2\beta + I_3 \cos^2\gamma) + \frac{1}{2} m l^2 \dot{\Omega}^2$$

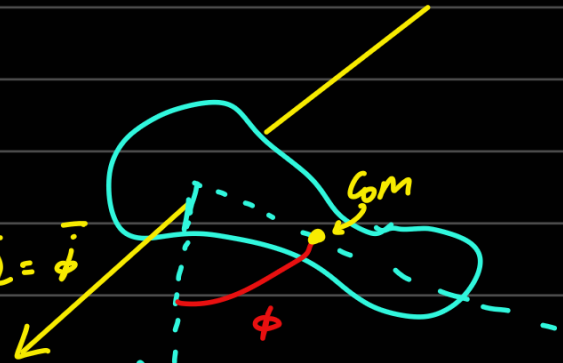
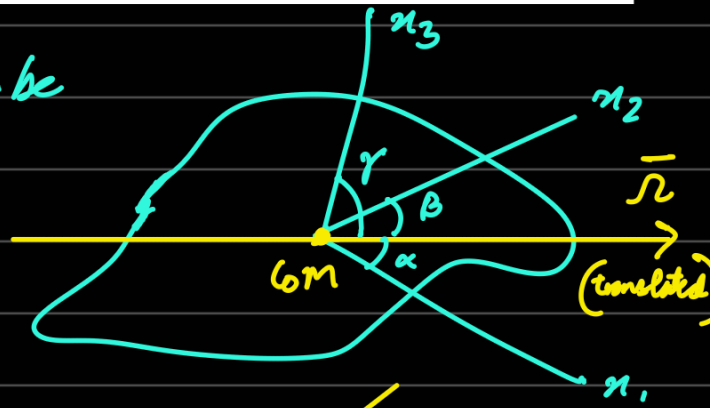
$$\mathcal{L} = \frac{\dot{\phi}^2}{2} (I_1 \cos^2\alpha + I_2 \cos^2\beta + I_3 \cos^2\gamma + m l^2) - mgl \frac{\phi^2}{2}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) = \frac{\partial \mathcal{L}}{\partial \phi}$$

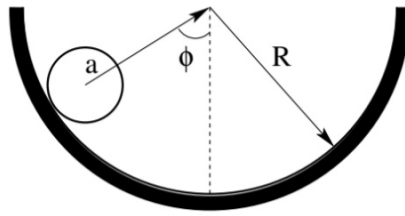
$$\Rightarrow \ddot{\phi} = -\omega^2 \phi \quad \text{with} \quad \omega^2 = \frac{mgl}{m l^2 + I_1 \cos^2\alpha + I_2 \cos^2\beta + I_3 \cos^2\gamma}$$

let's check this for a simple pendulum.

$$I_1 = I_2 = I_3 = 0 \quad \text{about CoM} \Rightarrow \omega^2 = g/l$$



3. Kinetic energy of a rolling cylinder: Obtain the kinetic energy of a homogeneous cylinder of radius a that is rolling inside a cylindrical surface of radius R as shown in the figure below



3

$$T_{\text{rot}} = \frac{1}{2} \left(\frac{ma^2}{2} \Omega^2 + 0 + 0 \right)$$

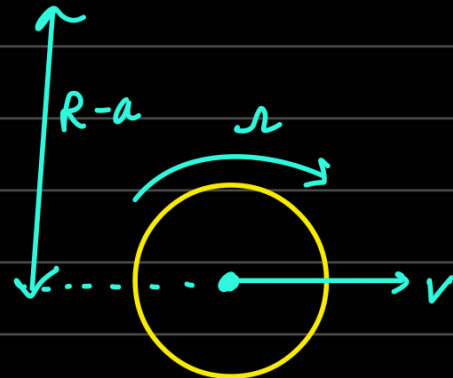
$$T_{\text{rot}} = \frac{ma^2 \Omega^2}{4}$$

$$T_{\text{trans.}} = \frac{mv^2}{2}$$

$$v = a\Omega = (R-a)\dot{\phi}$$

$$\Rightarrow \Omega = \left(\frac{R-a}{a} \right) \dot{\phi}$$

$$\therefore T = \frac{3}{4} ma^2 \Omega^2 = \frac{3}{4} m (R-a)^2 \dot{\phi}^2$$



4. Deflection of a freely falling body: Find the deflection of a freely falling body from the vertical caused by the Earth's rotation, assuming the angular velocity of this rotation to be small.

$$m \frac{d\vec{v}}{dt} = -\frac{\partial U}{\partial \vec{r}} - m\vec{w} + m\vec{r} \times \vec{\dot{\Omega}} + 2m\vec{v} \times \vec{\Omega} + m\vec{\Omega} \times (\vec{r} \times \vec{\Omega})$$

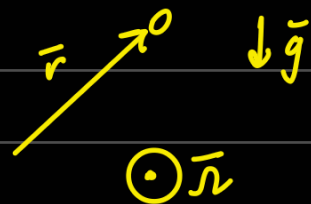
in a non-inertial frame

where $\vec{w} \rightarrow$ translational acceleration of CoM.

Here, $\vec{w} = 0$, $\vec{\Omega} \rightarrow$ constant, $U = -m\vec{g} \cdot \vec{r}$

$$m \frac{d\vec{v}}{dt} = m\vec{g} + 2m(\vec{v} \times \vec{\Omega}) + \cancel{0(\Omega^2)}$$

$$\frac{d\vec{v}}{dt} = \vec{g} + 2(\vec{v} \times \vec{\Omega})$$



let $\vec{v} = \vec{v}_1 + \vec{v}_2$ such that $\frac{d\vec{v}_1}{dt} = \vec{g}$, $\vec{v}_2 \ll \vec{v}_1$

$$\Rightarrow \cancel{\frac{d\vec{v}_1}{dt}} + \frac{d\vec{v}_2}{dt} = \cancel{\vec{g}} + 2(\vec{v}_1 \times \vec{\omega}) + 2(\vec{v}_2 \times \vec{\omega}) \approx 0$$

$$\frac{d\vec{v}_2}{dt} = 2(\vec{v}_1 \times \vec{\omega})$$

$$\text{Now } \vec{v}_1 = \vec{g}t + \vec{c}_0 \Rightarrow \vec{r}_1 = \vec{g}\frac{t^2}{2} + \vec{c}_0 t + \vec{c}_1$$

$$\Rightarrow \frac{d\vec{v}_2}{dt} = 2\left(\vec{g}t \times \vec{\omega} + \vec{c}_0 \times \vec{\omega}\right)$$

$$\Rightarrow \vec{v}_2 = 2\left(\vec{g} \times \vec{\omega} \frac{t^2}{2} + \vec{c}_0 \times \vec{\omega} t\right)$$

$$\Rightarrow \vec{r}_2 = \vec{g} \times \vec{\omega} \frac{t^3}{3} + \vec{c}_0 \times \vec{\omega} t^2 + \vec{c}_2$$

$$\therefore \vec{r} = \vec{r}_1 + \vec{r}_2 = (\vec{g} \times \vec{\omega}) \frac{t^3}{3} + \left((\vec{c}_0 \times \vec{\omega}) + \frac{\vec{g}}{2}\right) t^2 + \vec{c}_0 t + \vec{c}_1$$

$$\text{Now } \vec{r}(0) = \vec{H} = H\hat{y} \Rightarrow \vec{c}_1 = H\hat{y}$$

$$\vec{v}(0) = \vec{0} \Rightarrow \vec{c}_0 = \vec{0}$$

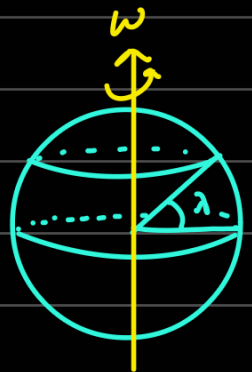
$$\therefore \vec{r} = (\vec{g} \times \vec{\omega}) \frac{t^3}{3} + \frac{\vec{g}}{2} t^2 + H\hat{y}$$

$$\vec{g} = -g\hat{y} \quad \vec{\omega} = \omega\hat{z} \Rightarrow \vec{g} \times \vec{\omega} = -g\omega\hat{x}$$

$$\therefore \vec{r} = -g\omega \frac{t^3}{3} \hat{x} + \left(H - \frac{gt^2}{2}\right) \hat{y}$$

$$\text{And deflection} = |\delta\vec{r}| = g\omega \frac{t^3}{3} \text{ where } t \approx \sqrt{\frac{2H}{g}}$$

$$|\delta\vec{r}| = \frac{g\omega}{3} \left(\frac{2H}{g}\right)^{\frac{3}{2}} \cos\lambda \quad \omega = \omega \sin\left(\frac{\pi}{2} - \lambda\right) = \omega \cos\lambda$$



5. Effects of Earth's rotation on a pendulum: Determine the effect of the Earth's rotation on small oscillations of a pendulum.

Note: This effect is best illustrated by the so-called Foucault's pendulum.

$$\mathcal{L} = \frac{1}{2} m \dot{\mathbf{r}}^2 + m \bar{\mathbf{v}} \cdot (\bar{\boldsymbol{\Omega}} \times \bar{\mathbf{r}}) + \frac{1}{2} m (\bar{\boldsymbol{\Omega}} \times \bar{\mathbf{r}})^2 - m \bar{\mathbf{v}} \cdot \bar{\mathbf{r}} - U$$

$$m \frac{d\bar{\mathbf{v}}}{dt} = - \frac{\partial U}{\partial \bar{\mathbf{r}}} + 2m(\bar{\mathbf{v}} \times \bar{\boldsymbol{\Omega}})$$

$$U = -mgL \frac{\phi^2}{2} = -\frac{mg}{2L} (x^2 + y^2) = -\frac{m\omega^2}{2} (x^2 + y^2)$$

$$\frac{\partial U}{\partial \bar{\mathbf{r}}} = -m\omega^2 x \hat{x} - m\omega^2 y \hat{y}$$

$$\bar{\boldsymbol{\Omega}} = \Omega_z \hat{z} \quad \bar{\mathbf{v}} =$$